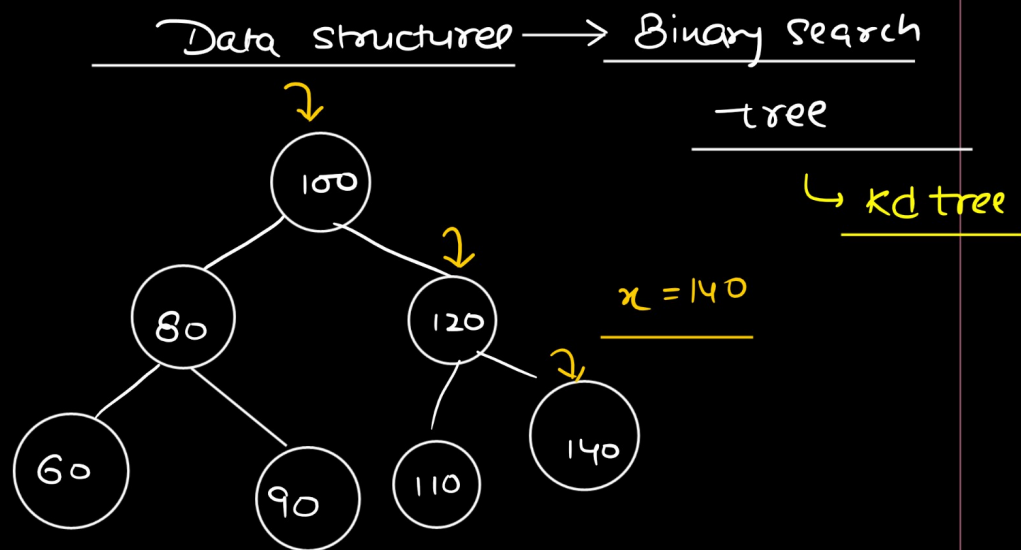
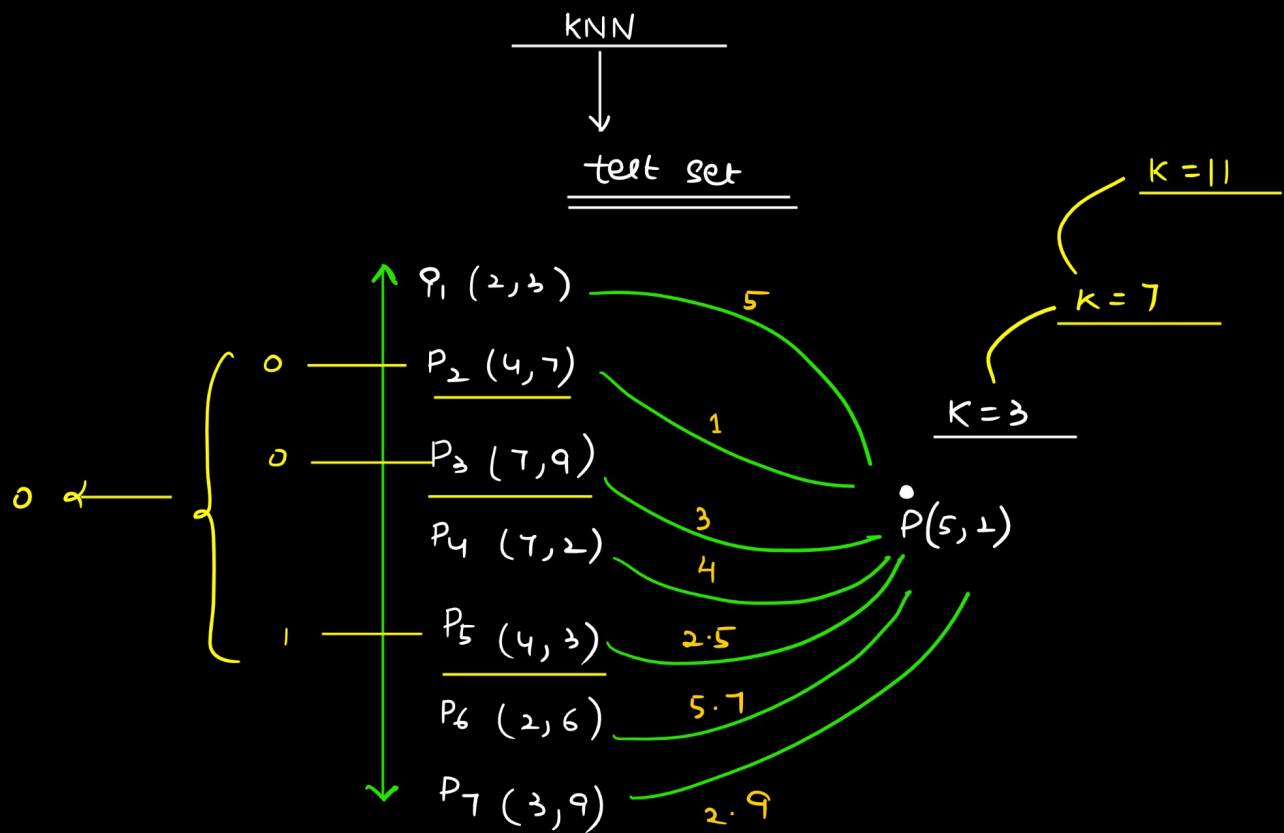
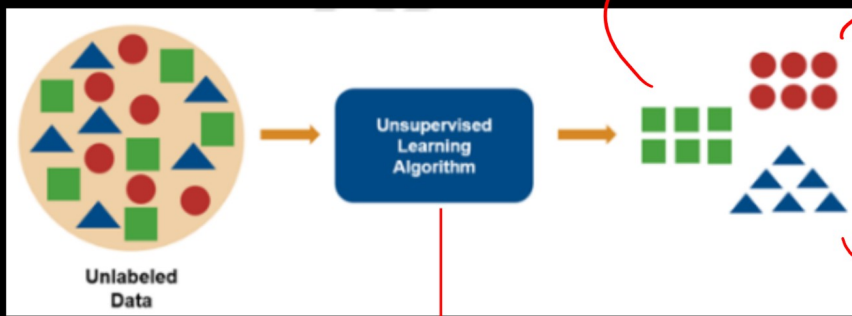
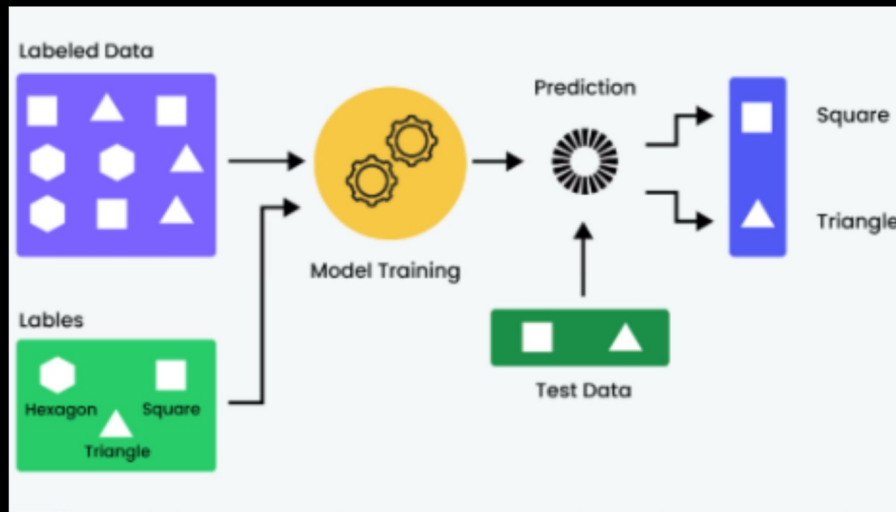




Unsupervised Machine Learning



EDA/
Data
preprocessing
Dimensionality
reduction
clustering
association
rule

Clustering algorithms

K-Means

→ Distance measure
(Euclidean
distance)

K-Means

↳ ① Maths

② Implementation

same cluster → highly similar

Different clusters → highly dissimilar

Real time scenarios

↳ customer representation

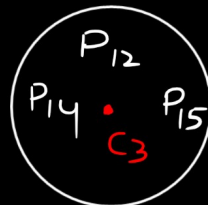
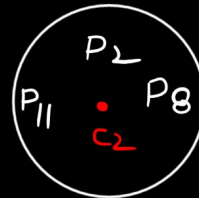
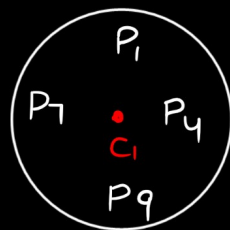
market basket analysis

image representation

→ DL

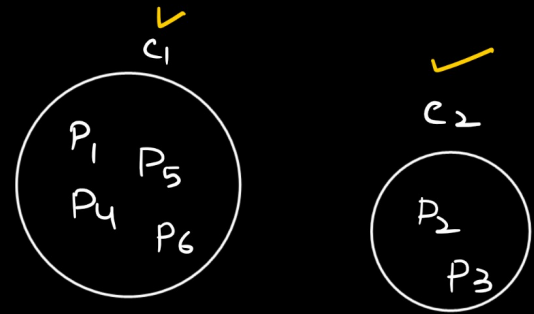
K-Means

→ k → num of Centroids



$k = 3$

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72			c1
2	170	56			c2
3	168	60	$\sqrt{433}$	$\sqrt{20}$	c2
4	179	68	$\sqrt{52}$	$\sqrt{225}$	c1
5	182	72	$\sqrt{9}$	$\sqrt{400}$	c1
6	188	77	$\sqrt{34}$	$\sqrt{765}$	c1



Iteration 2

until we get stabilized centroid

→ Re-evaluate the centroid

$c_1 \Rightarrow \text{average}(x_i, y_i)$

num of clusters c_2

$k = 3$

→ randomly initialize

the 3

Centroids

↳ nearby

K-means++

→ never be very

close to

each other (centroids)
customers

Age

Monthly spending

$k = 3$

✓ young low spenders

✓ middle-age high spenders

targeted

Marketing

premium

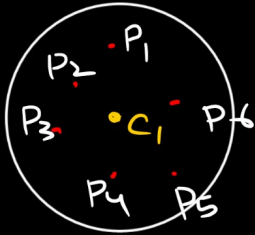
customers

Objective function

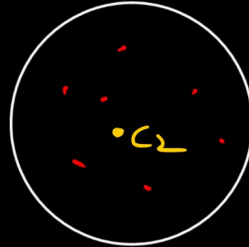
$$\text{WCSS} = S_1 + S_2 + S_3$$

K-Means $\rightarrow$ minimize WCSS

$$S_1 = \sum \|C_1 - x_i\|^2$$



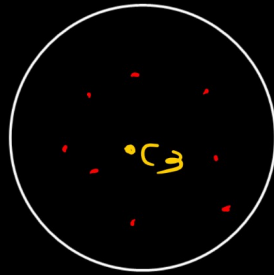
$$S_2 = \sum \|C_2 - x_i\|^2$$



Within
Cluster

Sum of

Square



$$S_3 = \sum \|C_3 - x_i\|^2$$

How to get best value of k ?

Elbow Method

